Daniel Foster

LinkedIn | Canada & EU Citizen | Bilingual (English & French) | UK Visa

Summary

- Master of Applied Science candidate in Aerospace Engineering, thesis in collaboration with Bombardier
- Industry experience in CFD at Pratt & Whitney Canada, undergraduate thesis in CFD code development
- Strong communication skills, fully bilingual in English and French

Education

Master of Applied Science in Aerospace Engineering — University of Toronto
GPA: 4.0/4.0 — 2024 - 2026 (August)
Thesis: Noise Generation Mechanisms of Slat Tracks and Their Mitigation – in collaboration with Bombardier Inc.
Herbert Spencer Ribner Memorial Scholarship (2024)
Relevant coursework: Advanced Flight Dynamics (A), Fundamentals of CFD (A+), Special Topics in CFD (A+)

Bachelor of Applied Science in Engineering Science (Honours), Aerospace Engineering — University of Toronto
GPA: 3.6/4.0 — 2019 - 2024
Thesis: Dissipation-Based Homotopy Continuation Methods for Steady Aerodynamic Flows
Relevant coursework: Aerodynamics (A+), Computational Structural Mechanics & Design Optimization (A)

Projects

CFD Solver Development
Numerical Methods for Convergence Acceleration of Aerodynamic Flow Simulations

- Developed and implemented a homotopy continuation method within research CFD code to improve robustness of steady compressible Navier-Stokes simulations
- Investigated nonlinear solver behaviour, Newton-Krylov-Schwarz methods, convergence acceleration techniques, and numerical continuation strategies
- Modified a large-scale C++ CFD codebase and executed verification studies using HPC resources
- Evaluated solver performance against existing continuation methods and assessed impacts on convergence

Experience

University of Toronto Institute for Aerospace Studies — North York, ON, Canada
Experimental Fluid Dynamics – Graduate Researcher — Sep. 2024 - Present

- Led wind-tunnel test programs for experimental quantification of aerodynamic noise generated by high-lift slat systems, in collaboration with Bombardier
- Programmed automated data analysis tools and scripts using Python and MATLAB
- Coordinated experimental hardware, setup activities, and test execution with engineering stakeholders to translate technical requirements into robust test programs
- Presented reports and communicated results to Acoustics & Vibration engineering stakeholders at Bombardier

Pratt & Whitney Canada — Mississauga, ON, Canada
Compressor Aerodynamics – CFD Integrator (Aerothermal Fluids Co-op Student) — May 2022 - Sep. 2023

- Led CFD performance benchmarking studies on computational aircraft engine models, in support of in-house methods and tools development within Compressor Aerodynamics
- Generated aerodynamic solutions using CFD software to produce estimates of pressure distributions and aerodynamic loads, validating results with experimental data
- Developed complex CFD workflows for aircraft engine designs from meshing to post-processing, contributing to development of in-house CFD tools and validating new CFD capabilities
- Presented reports and communicated results to Compressor Aerodynamics engineering stakeholders at Pratt & Whitney Canada

Skills

CFD: Finite-difference & finite-volume algorithms, multigrid, high-resolution schemes, ANSYS, STAR-CCM+
Programming: Python, C++, MATLAB, UNIX/LINUX environments, HPC best practices
Testing: CAD (SOLIDWORKS), wind tunnel testing
Languages: English (native language), French (professional working proficiency – DELF B2)